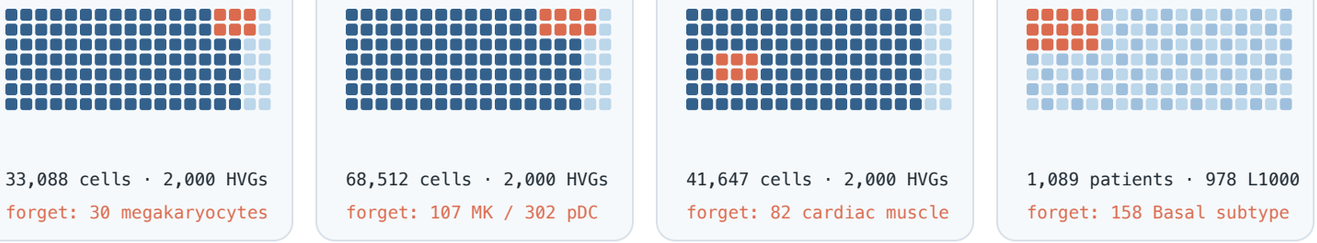
A Datasets

Four cohorts, two modalities. Each structured forget set is a rare biologically coherent subpopulation.

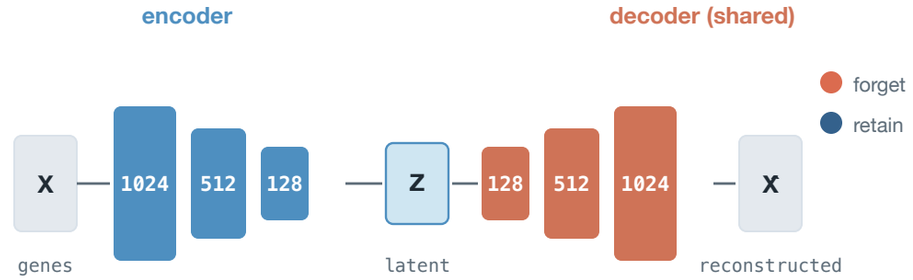


retain cells · unseen (matched negatives) · forget set (structured)

train

B VAE

scVI-style encoder-decoder with shared output parameters.
Latent dim 32.

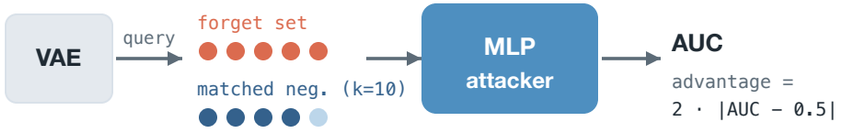


$$L = -E[\log p(x|z)] + KL[q(z|x) \parallel N(0,I)]$$

both pass through the same parameters

C Membership inference

One MLP attacker on 70 model-internal features, trained on the baseline and applied to every model; a likelihood audit vs. retrained models.



PBMC cluster 13, attacker AUC

baseline	retrained	chance
0.80	0.53	0.50

D Unlearning & evaluation

Nine post-hoc methods, two constructive approaches, a fine-tune with a likelihood target, and DP-SGD, judged against models retrained without the forget set.

METHODS, GROUPED BY WHAT THEY DO

Edit, then fine-tune	retain FT · GA · Fisher · SSD · SCRUB · Moon
Continuous objective	extra-grad. · contrastive · frozen critics
Constructive	synthetic aug. · repr. alignment
Targeted	fine-tune to a likelihood target
From scratch	DP-SGD · retrained models

